

Week 3 of CADD Training

FASTA RETRIEVAL • ALPHAFOLD3 • STRUCTURAL ALIGNMENT

Type	Copies	Sequence			
Protein	1	10 GPDEEPEEPG	20 RRGSFVEMVD	30 NLRGKSGQGY	40 YVEMTVGSPP
		70 HPFLHRYYQR	80 QLSSTYRDLR	90 KGVVVPYTQG	160 KWEGELGTDL
		130 SDKFFINGSN	140 WEGILGLAYA	150 EIARPDDSLE	160 PFFDSLVKQT
		190 EVLASVGGSM	200 IIGGIDHSLY	210 TGSLWYTPIR	220 REWYYEVIIV
		250 SIVDSGTTNL	260 RLPKKVFEAA	270 VKSIKAASST	280 EKFPDGFVLG
		310 LYLMGEVTNG	380 SFRITILPQQ	330 VLRPVEDVAT	340 SQDDCYKFAI
		370 FDRARKRIGF	380 AVSACHVHDE	390 FRTAAVEGPF	400 VTLDMEDCGY
Seed: 499171687					



01

Every structure
starts with a
sequence.

Protein ID: 1U72

FASTA Sequence: rcsb_pdb_1U72

Source: bioinformatic Identifier: rcsb_pdb_1U72 Organism: Homo sap...

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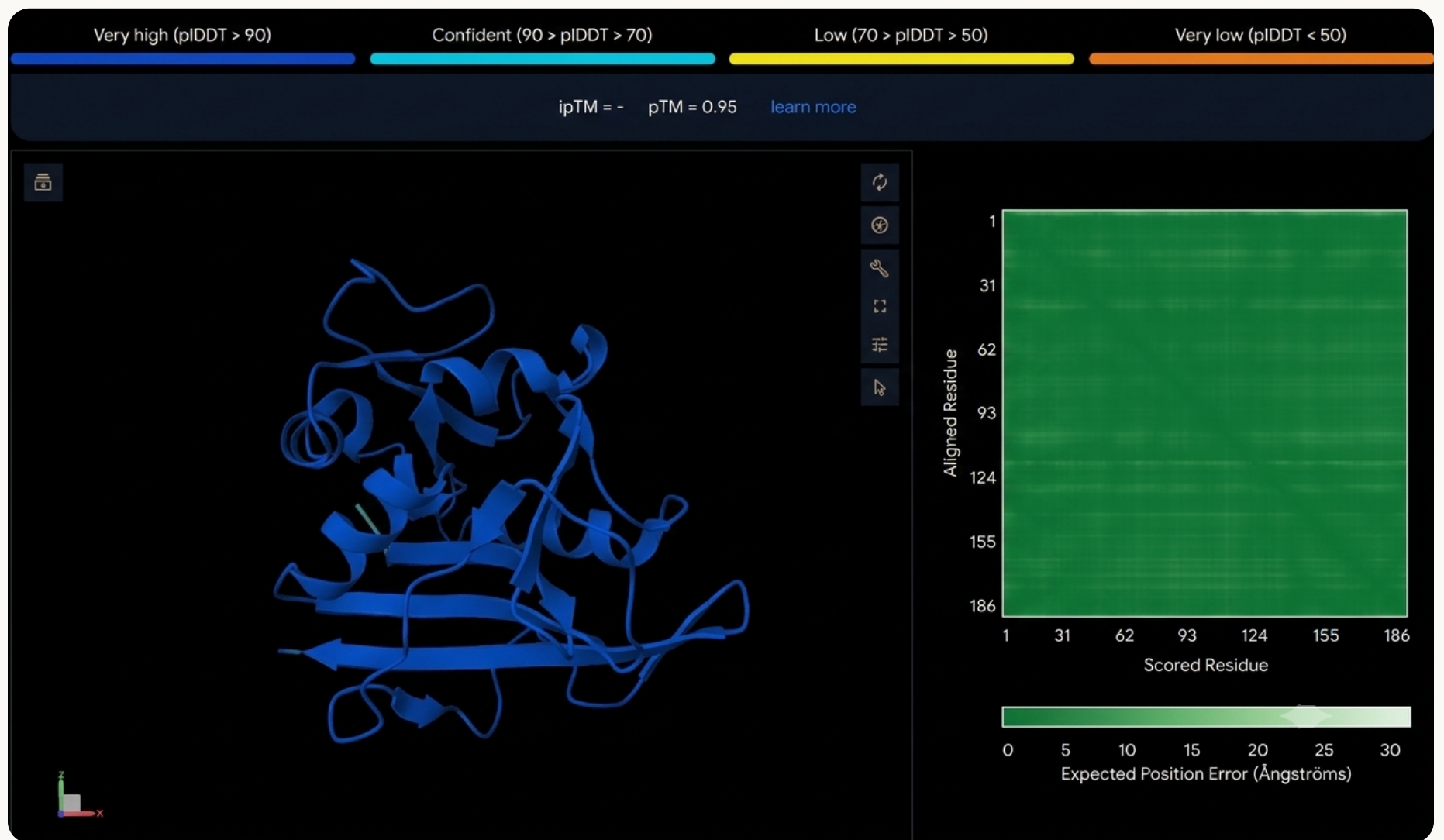
```
>1U72_1|Chain A|Dihydrofolate reductase|Homo sapiens (9606)  
VGSLNCIVAVSQNMGIGKNNGDLPWPPLRNEFRYFQRMTTSSVEGQKQLVIMGKKTWFSIPEK  
NRPLKGRINLVLSRELKEPPQGAHFLSRSLDDALKKLTEQPELAANKVDMVWIVGGSSVYKE.
```

SEQUENCE (FASTA) → ALPHAFOLD3 → 3D STRUCTURE

Retrieved the FASTA sequence of 1U72 (Human Dihydrofolate Reductase) from the Protein Data Bank and used it as the starting input for structure prediction.

02

AI Converts Sequence into Structure

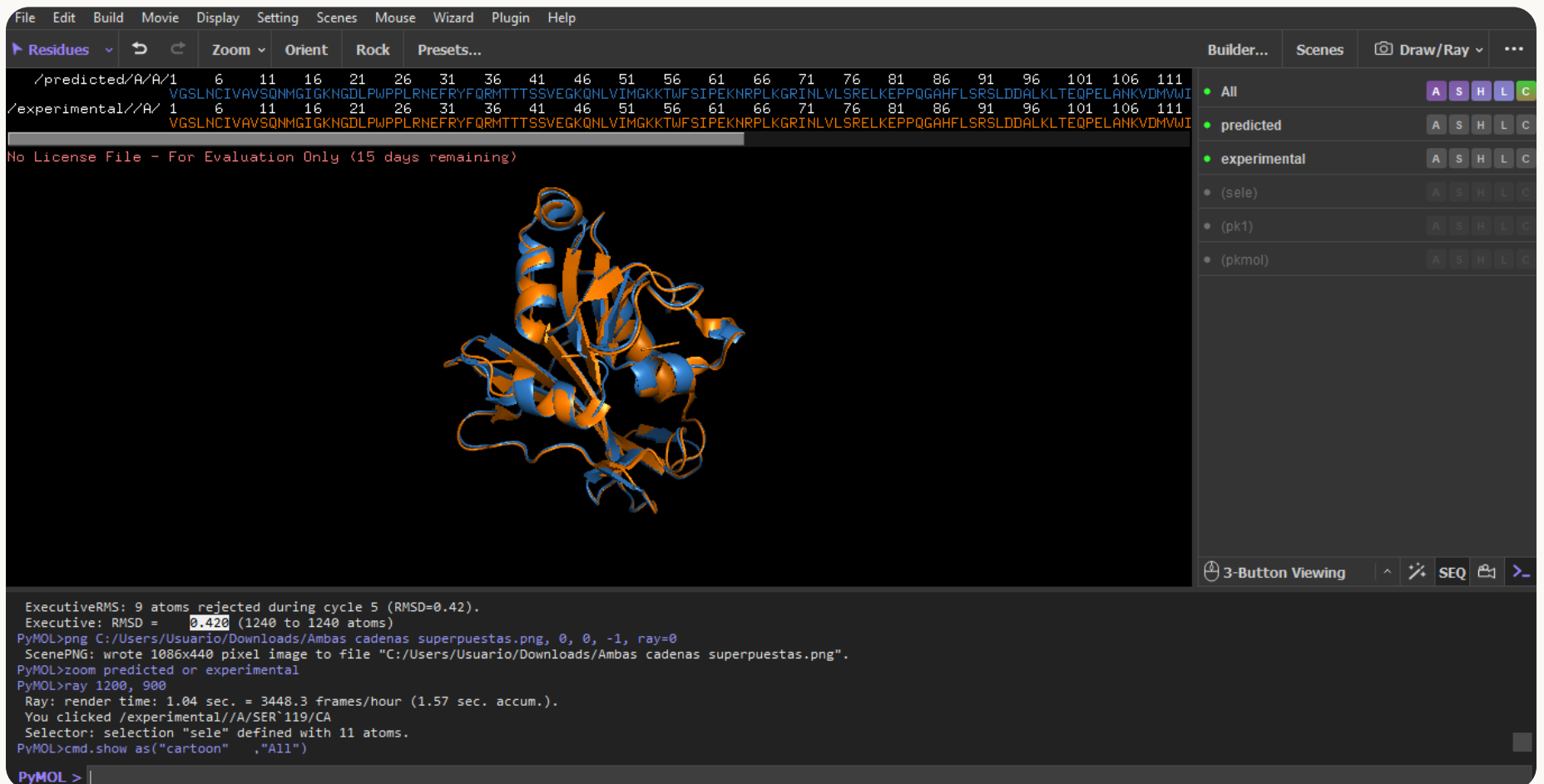


Predicted 3D Model

Generated a predicted structural model of 1U72 using AlphaFold3 from the retrieved FASTA sequence. Model confidence: pTM = 0.95, with most of the structure falling in the very high confidence range (pLDDT > 90).

03

Prediction vs Reality



Aligned Structures

Aligned the predicted 1U72 model (blue) with the cleaned experimental structure (orange) in PyMOL to evaluate structural similarity.

Excellent structural agreement. RMSD = 0.420 Å

What this means:

< 1.0 Å = Excellent agreement < 2.0 Å = Good match > 3.0 Å = Significant differences